

SILAS V. BLACKWOOD

Chief Technology Officer | Principal .NET Enterprise Architect & Global Engineering Leader

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PROFESSIONAL SUMMARY

Visionary Technology Executive and Principal .NET Architect with over 12 years of high-impact experience driving large-scale digital transformations for Fortune 100 enterprises and high-growth technology firms. Distinguished for architecting the "Core-Sync 2.0" global settlement engine, which processes over \$15B in daily clearing volume with deterministic sub-10ms latency. Silas is an elite technical leader who has successfully managed engineering organizations of 150+ professionals, bridging the gap between low-level CLR optimization (memory management, SIMD intrinsics) and high-level strategic business growth.

Expert in the deployment of modern .NET 8/9 ecosystems, high-density Kubernetes orchestration, and globally distributed event-driven architectures. A strategic architect who has achieved a consistent 45% reduction in infrastructure costs while increasing developer velocity by 300% through the implementation of automated "IDP" (Internal Developer Platforms). Proven mastery in navigating highly regulated sectors including Fintech (PCI-DSS 4.0), Healthtech (HIPAA), and Defense (ITAR). His leadership philosophy centers on "Architectural Integrity," technical mentorship of high-potential staff, and the relentless pursuit of engineering excellence through evidence-based decision making and automated quality assurance at scale.

CORE COMPETENCIES

- **Enterprise Architecture:** Distributed Systems Design, Cloud-Native Microservices, Hexagonal/Clean Architecture, Domain-Driven Design (DDD), SAGA Patterns, CQRS, Event Sourcing.
- **.NET Ecosystem Mastery:** C# 12/13, ASP.NET Core 8/9, Entity Framework Core (Source Generators), SignalR (Real-time Hubs), gRPC (High-performance APIs), Blazor WASM.
- **Cloud Strategy & Infrastructure:** Microsoft Azure (AKS, Functions, Cosmos DB, Service Bus, Key Vault), AWS (EKS, Lambda, Aurora), Multi-Cloud Hybrid Architectures.
- **Engineering Leadership:** OKR Strategic Planning, P&L Responsibility (\$180M+), Talent Acquisition & Leveling, Vendor Management, M&A Technical Due Diligence.
- **DevOps & Automation:** CI/CD Pipeline Architecture (Azure DevOps, GitHub Actions), Infrastructure-as-Code (Terraform, Bicep), KEDA for Event-driven Scaling.
- **Data Engineering:** SQL Server 2022 Performance Tuning, PostgreSQL (Citius Sharding), MongoDB, Cassandra, Redis Cluster Optimization, EventStoreDB.
- **Security & Governance:** Zero-Trust Architecture, OAuth2/OIDC Implementation, JWT Security, mTLS, OWASP Top 10 Mitigation, HIPAA, PCI-DSS, GDPR.
- **Quality Assurance:** Test-Driven Development (TDD), Consumer-Driven Contract Testing, Chaos Engineering, Playwright, BenchmarkDotNet.

TECHNICAL SKILLS

Programming Languages & Runtimes

- **Expert Level:** C# (.NET 8/9), TypeScript, Go (Golang), T-SQL, PowerShell.
- **Advanced Level:** F# (Functional domain modeling), JavaScript (ESNext), Python (ML Integration), Rust (FFI/Low-level modules), Bash.
- **Runtime Mastery:** CLR (Garbage Collector tuning, JIT optimization), Node.js, WebAssembly (WASM).

Microsoft .NET Stack

- **Web & API:** ASP.NET Core (Minimal APIs, SignalR, Web API, Identity), Blazor (WASM & Server), Microsoft Orleans (Virtual Actors).
- **Data Access:** Entity Framework Core (Source Generators, Compiled Queries), Dapper, AutoMapper, FluentValidation.
- **Messaging:** MassTransit (Azure/RabbitMQ/Kafka Transports), NServiceBus, SignalR Backplanes, Azure Service Bus.
- **Utilities:** MediatR, Serilog (Structured Logging), Polly (Resilience), xUnit, Moq, Bogus, BenchmarkDotNet.

Cloud & Infrastructure Architecture

- **Azure Core:** Azure Kubernetes Service (AKS), Container Apps, Azure Functions (Serverless), Logic Apps, API Management (APIM).
- **Network & Security:** Azure Front Door, WAF, Application Gateway, Azure Firewall, mTLS, Azure Key Vault, Managed Identities.
- **Automation:** Terraform (HCL), Bicep, ARM Templates, Pulumi, Helm Charts, Kustomize, Ansible.
- **CI/CD:** GitHub Actions (Reusable workflows), Azure DevOps (YAML Pipelines), Octopus Deploy, ArgoCD (GitOps).

Database & Analytics Ecosystem

- **Relational:** SQL Server 2022 (In-Memory OLTP, Temporal Tables), Azure SQL DB, PostgreSQL (PostGIS), MariaDB.
- **NoSQL & Specialty:** Cosmos DB (SQL/Mongo/Gremlin APIs), MongoDB, Apache Cassandra, Redis (Enterprise Cluster), Elasticsearch/ELK Stack, Pinecone (Vector DB).
- **Data Ingestion:** Azure Data Factory, Kafka Connect, Spark (.NET for Apache Spark), RabbitMQ.

Monitoring, Observability & Standards

- **Observability:** Application Insights, Datadog (APM), Prometheus, Grafana, OpenTelemetry, Splunk, Jaeger.
- **Standards:** REST, GraphQL (HotChocolate), gRPC, SOAP (Legacy support), HL7/FHIR, ISO 20022.
- **Design:** UML, C4 Modeling, Mermaid.js, Lucidchart, Architecture Decision Records (ADR).

PROFESSIONAL EXPERIENCE

Chief Technology Officer | Aetheris Financial Solutions | 2021 – Present

Seattle, WA / Global (Fintech Global Clearing & Settlement | Annual Revenue: \$5.8B)

Strategic Mandate & Executive Oversight: Recruited as the executive technical leader to oversee the modernization of the global clearinghouse infrastructure. Manage a global engineering organization of 150+ professionals across 4 continents. Directly responsible for a \$185M annual R&D budget. Report directly to the CEO and Board of Directors on technical risk, scalability projections, and cyber-resilience.

Key Measurable Achievements:

- **Global Settlement Transformation:** Architected a distributed .NET 8 settlement engine utilizing the SAGA pattern and Kafka. Replaced a legacy mainframe system, resulting in a 500% increase in transaction capacity and a reduction in settlement time from T+2 to real-time.
- **Cloud-Native Infrastructure:** Directed the migration of 450+ services to a multi-region Azure Kubernetes Service (AKS) environment. Implemented KEDA for event-driven autoscaling, reducing idle compute costs by \$12M annually.
- **High-Frequency Performance:** Optimized C# codebases using 'Span<T>', 'Memory<T>', and 'System.Runtime.Intrinsics'. Achieved a p99 latency of <5ms for high-frequency trading data ingestion, outperforming the previous Java-based solution by 40%.
- **Zero-Trust Security Implementation:** Led the global rollout of a mTLS and Identity-based security model across all internal microservices. Utilized Azure AD B2C and IdentityServer6 to secure 25M+ end-user identities, achieving zero security breaches in 3 years.
- **Productivity Engineering:** Spearheaded the development of an Internal Developer Platform (IDP) utilizing Backstage and custom C# CLI tools. Reduced the "Lead Time to Production" for new features from 4 weeks to 2 days.
- **Board-Level Advisory:** Successfully defended three multi-year technical roadmaps to the Board, securing an additional \$45M in funding for "AI-Assisted Fraud Detection" initiatives utilizing ML.NET and Azure Cognitive Services.
- **Talent Acquisition & Mentorship:** Re-architected the technical career ladder, introducing "Distinguished Engineer" levels. Reduced engineering turnover from 18% to 4% through a culture of "Radical Ownership" and technical empowerment.

VP of Engineering & Principal Architect | Novus Health Systems | 2017 – 2021

San Francisco, CA (Healthtech Data Interoperability | 20M+ Patients)

Operational Leadership: Served as the primary technical authority for the "Patient-360" data fabric. Directed 60 senior engineers and 20 data scientists in building a real-time clinical data exchange platform. Accountable for 100% HIPAA compliance across all cloud systems.

Key Measurable Achievements:

- **Clinical Data Mesh:** Architected a C#-based FHIR (Fast Healthcare Interoperability Resources) data mesh that integrated 500+ disparate EMR systems. Enabled real-time patient data availability for clinicians, reducing diagnostic delays by 25%.
- **Real-Time Telemetry Engine:** Built a high-frequency ingestion engine using ASP.NET Core SignalR and Azure Functions. Processed 3.5 billion telemetry points daily from wearable med-

ical devices with 99.999% reliability.

- **Database Scalability Optimization:** Directed the transition from a monolithic SQL Server instance to a sharded PostgreSQL architecture using Citus and .NET Entity Framework Core. Improved data retrieval performance for patient histories by 400% during peak usage.
- **Regulatory Navigation:** Acted as the lead technical witness for 4 federal healthcare audits. Implemented automated data-anonymization pipelines that ensured 100% compliance with state and federal privacy laws.
- **Chaos Engineering Deployment:** Introduced Gremlin and custom .NET chaos experiments into the CI/CD pipeline. Reduced production major incidents by 65% by proactively identifying distributed system failures in staging.
- **Executive Recruitment:** Personally hired 40+ senior and staff level developers, focusing on a "T-shaped" skill set (deep .NET expertise with broad cloud infrastructure knowledge).

Director of Software Development | Velocity Retail Group | 2014 – 2017

Charlotte, NC (E-commerce Logistics & Global Supply Chain)

Role Overview: Primary leader for the "Omni-Channel Logistics" engineering division. Focused on high-volume inventory forecasting and real-time order routing for a global retailer.

Key Measurable Achievements:

- **Inventory Forecasting Revolution:** Developed a multi-threaded C# forecasting engine utilizing the Task Parallel Library (TPL) and PLINQ. Reduced the daily inventory reconciliation window from 14 hours to 30 minutes for 5,000+ retail locations.
- **Micro-frontend Transition:** Led the technical move from a legacy ASP.NET Web Forms dashboard to a micro-frontend architecture using Blazor WebAssembly and C#. Enabled 5 different product teams to deploy updates independently without downtime.
- **Messaging Reliability Overhaul:** Re-engineered the order-routing system using RabbitMQ and MassTransit with "Guaranteed Delivery" patterns. Eliminated 100% of data loss during high-volume events like Black Friday.
- **Cloud Migration Lead:** Directed the migration of the primary logistics platform from a private data center to Microsoft Azure. Reduced annual physical infrastructure spend by \$9.5M while improving disaster recovery RTO from 12 hours to 10 minutes.
- **Continuous Quality Initiative:** Introduced mandatory "Property-Based Testing" and TDD for all core business modules. Increased code coverage from 20% to 88%, resulting in a 40% reduction in production hotfixes.

Senior .NET Developer / Architect | CoreLogix SaaS | 2011 – 2014

San Antonio, TX (Enterprise Resource Planning - ERP Solutions)

Key Measurable Achievements:

- **Legacy Modernization Project:** Acted as the lead developer for the modernization of a legacy ASP.NET 4.0 application to ASP.NET MVC 5. Implementation of the "Repository Pattern" and "Unit of Work" decoupled the business logic from the data layer, reducing maintenance effort by 35%.
- **API Ecosystem Development:** Built a suite of RESTful APIs for mobile application integration. Implemented custom OAuth2 filters and response caching, resulting in a 50% improvement in mobile app responsiveness.
- **Unit Testing Advocate:** Introduced the first automated unit testing framework (NUnit) to the organization. Developed a library of reusable "test-harnesses" that reduced the time to create new integration tests by 60%.
- **Performance Tuning:** Redesigned a critical financial reporting module using Dapper and raw SQL instead of heavy ORM calls, reducing report generation time for large enterprise clients from 5 minutes to 15 seconds.

KEY PROJECTS PORTFOLIO

Project Hydra: The Global Ledger Resilience Initiative

Role: Executive Lead & Chief Architect | **Duration:** 24 Months

Complexity: Transitioned a COBOL-based mainframe banking core to a modern .NET 9 distributed ledger for a Tier-1 global bank.

- **Objective:** Modernize the clearing and settlement process to support 24/7 global operations and eliminate the "End-of-Day" processing bottleneck.
- **Strategy:** Utilized the "Strangler Fig" pattern to incrementally migrate monolithic components. Implemented a custom "Event-Bridge" in C# to sync data between legacy and modern systems in real-time.
- **Implementation:** Built a distributed actor system using Microsoft Orleans to manage individual account states. Leveraged Cosmos DB with multi-region write capability to ensure sub-second global data consistency.
- **Outcome:** Successfully migrated 500M+ accounts without a single localized outage. Achieved sub-second reconciliation for global inter-bank transfers, saving an estimated \$200M in annual operational risk.

Project Sentinel AI: Real-Time Fraud Detection Fabric

Role: Technical Strategy Lead | **Duration:** 18 Months

Objective: Build an AI-driven fraud detection engine that analyzes 100,000 transactions per second in real-time.

- **Strategy:** Integrated ML.NET models directly into the high-frequency telemetry pipeline. Developed a hybrid cloud-edge architecture where initial filtering occurs in a .NET-based Edge worker (K3s).
- **Action:** Developed a custom "Streaming Inference Engine" in C# that utilizes SIMD (Single Instruction, Multiple Data) intrinsics for ultra-fast pattern matching against historical fraud vectors.
- **Outcome:** Reduced "False Positive" fraud alerts by 45%, significantly improving customer satisfaction scores. Prevented \$85M in fraudulent transactions in the first 12 months of deployment.

Project Vertex Mesh: Enterprise Service Mesh Implementation

Role: Principal Architect | **Duration:** 12 Months

Objective: Standardize observability, security, and traffic management for 800+ microservices.

- **Strategy:** Deployed Istio Service Mesh on Azure Kubernetes Service (AKS). Built a custom .NET "Sidecar Utility" that automated mTLS certificate rotation and health check verification.
- **Outcome:** Achieved 100% visibility into inter-service communication. Enabled "Canary Deployments" and "Blue-Green" switching for all product teams, reducing the risk of bad releases to near zero.

Project Green-Build: Sustainable Cloud Automation

Role: Engineering Director | **Outcome:** Saved \$15M in annual cloud spend.

- **Action:** Developed an automated "Cloud Sweeper" in C# that identifies under-utilized resources across Azure and AWS. Integrated with the CI/CD pipeline to automatically "right-size" container resource limits based on historical usage data.

LEADERSHIP & TEAM MANAGEMENT

- **Scaling Engineering Organizations:** Successfully grew the .NET engineering organization at Aetheris from 30 to 150+ professionals across 4 global technical hubs (Seattle, London, Berlin, Bangalore).
- **The "Blackwood" Engineering Standard:** Authored a 250-page internal standard for .NET development, covering everything from asynchronous programming best practices to standardized C# coding styles and Entity Framework Core performance guidelines.
- **OKR-Driven Strategy:** Implemented a "Value-Stream" focused OKR framework that aligned engineering outputs with direct commercial revenue targets, resulting in a 40% increase in feature velocity.
- **Crisis Management Leadership:** Managed the technical response to a major industry-wide zero-day vulnerability in 2022. Successfully patched 6,000+ production services within 4 hours with zero data exposure.
- **Executive Presence:** Provide bi-weekly technical briefings to the CEO, CTO, and Board members on system health, risk mitigation, and the ROI of technical debt remediation.
- **Diversity & Inclusion in Tech:** Spearheaded a "Women in Engineering" recruitment initiative, increasing female representation in the technical organization by 45% over 3 years.
- **Cultural Transformation:** Introduced a "DevSecOps" mindset where security is integrated into the earliest stages of the SDLC, reducing security-related release delays by 70%.

PROCESS IMPROVEMENTS & AUTOMATION

- **Automated Load Testing Platform:** Built a platform using k6 and Azure Container Instances that automatically runs load tests for every major release. Identified 30+ performance regressions before they reached production, saving an estimated \$8M in potential downtime.
- **RPA for Financial Reconciliation:** Developed a C#-based robotic process automation tool to handle manual data entry for international currency adjustments. Reduced human error by 99% and saved 600 man-hours monthly.
- **Self-Healing Cloud Operators:** Built a custom Kubernetes Operator in Go/C# that monitors microservice health and automatically adjusts memory limits and connection pools based on real-time traffic anomalies.
- **Automated Compliance Documentation:** Created a tool that automatically generates HIPAA and SOC2 compliance evidence from Azure DevOps logs and GitHub audit trails. Reduced manual audit prep time by 90%.
- **Shadow Traffic Deployment:** Implemented a "Dark Launch" proxy that duplicates production traffic to a testing environment, allowing for the verification of performance and logic on live data without impacting users.
- **Centralized Log Aggregation:** Standardized logging via a custom NuGet package that automatically enriches logs with trace IDs, user context, and machine data, reducing troubleshooting time by 60%.

STRATEGIC VISION: THE FUTURE OF .NET ENGINEERING

The Era of AI-Augmented Development

I believe the next decade of software engineering will be defined by the transition from "Manual Coding" to "AI-Augmented Architecture." I am positioning my teams to use AI (GitHub Copilot, Azure OpenAI) not just for code completion, but for automated unit testing, architectural pattern detection, and proactive security hardening. My mission is to build the infrastructure that allows developers to focus on high-value business logic while AI handles the "Cognitive Drudgery" of the SDLC.

Sustainability in Software

In an age of increasing environmental awareness, "Green Coding" is a commercial necessity. My strategy involves leveraging .NET 9's performance enhancements and container density to reduce the carbon footprint of our cloud infrastructure. By optimizing the code-to-compute ratio, we improve both corporate sustainability and shareholder value.

Data Sovereignty & Multi-Cloud Resiliency

As geopolitical volatility increases, technical resiliency must include data sovereignty. I champion a "Cloud-Agnostic" .NET architecture where the business logic is strictly separated from the underlying cloud provider via the Sidecar and Service Mesh patterns. This ensures that our platforms can pivot between Azure, AWS, or local data centers with minimal downtime.

EDUCATION

Master of Science in Software Engineering & Cloud Architecture

Massachusetts Institute of Technology (MIT), Cambridge, MA

- Focus Areas: Reliability Engineering, Distributed Computing, Statistical Defect Prediction.
- Honors: MIT Presidential Fellowship for Excellence in Research.
- Thesis: "Dynamic Resource Allocation in Large-Scale .NET Microservices."

Bachelor of Science in Computer Science & Mathematics

The University of Washington, Seattle, WA

- Honors: Summa Cum Laude; President of the UW Computer Science Honor Society.
- Recipient of the Engineering Dean's Excellence Scholarship.

CERTIFICATIONS

- **Microsoft Certified: Azure Solutions Architect Expert**
- **Microsoft Certified: DevOps Engineer Expert**
- **CKA:** Certified Kubernetes Administrator (Cloud Native Computing Foundation)
- **CISSP:** Certified Information Systems Security Professional (ISC2)
- **ISTQB-CTEL:** Test Automation Strategy Specialist
- **Lean Six Sigma Black Belt** (Process Optimization focus)

- **Pragmatic Institute:** Certified Product Manager (Level VI)

TOOLS & TECHNOLOGIES REFERENCE

IDEs & Dev Tools: Visual Studio 2022 Enterprise, VS Code, JetBrains Rider, JProfiler, Benchmark-DotNet.

Core Frameworks: .NET 8/9, .NET Core 3.1, .NET Framework 4.8, Blazor, SignalR, gRPC, ML.NET, Maui.

Cloud & DevOps: Azure Kubernetes Service (AKS), Azure DevOps, Terraform, Bicep, Helm, ArgoCD, Jenkins.

Persistence: SQL Server, Cosmos DB, PostgreSQL, MongoDB, ScyllaDB, Redis, Elasticsearch.

Messaging: Azure Service Bus, Kafka, RabbitMQ, MassTransit, NServiceBus, Google Pub/Sub.

Observability: Application Insights, Datadog, Grafana, Prometheus, OpenTelemetry, ELK Stack.

Compliance: Vanta, Drata, OneTrust, Burp Suite, Snyk, SonarQube, Prisma Cloud.

PUBLICATIONS & CONFERENCES

Selected Speaking Engagements

- **Keynote Speaker:** "The Death of the Monolith: Scaling .NET into the 2030s," *.NET Conf 2023*.
- **Featured Speaker:** "High-Frequency .NET: Lessons from a \$15B FinTech Engine," *Microsoft Ignite 2022*.
- **Panelist:** "Microservices vs. Monolith: The Practical Middle Ground," *QCon San Francisco 2021*.
- **Workshop Lead:** "Performance Tuning the CLR for Low Latency," *NDC London 2020*.
- **Guest Lecturer:** "Advanced Distributed Systems," *University of Washington - McCombs School of Business*.

Selected Peer-Reviewed Publications

- **"Predictive Resource Scaling in Kubernetes using Machine Learning and .NET"**
IEEE Journal of Software Systems, 2023.
- **"Beyond REST: Comparing gRPC and RSocket for High-Frequency C# Services"**
The Journal of High Performance Systems, 2022.
- **"The Cost of Complexity: Quantifying Architectural Debt in Enterprise .NET Systems"**
International Software Engineering Journal, 2021.
- **"Shift-Left Security: Automating Vulnerability Detection in .NET CI/CD Pipelines"**
Cybersecurity Digest, 2020.

LANGUAGES & GLOBAL EXPOSURE

- **English:** Native Proficiency.
- **German:** Professional Working Proficiency (C1).
- **Spanish:** Conversational.
- **Technical Contexts:** Expert in navigating international technical cultures and labor laws for engineering recruitment in the EU, APAC, and North America.
- **Global Footprint:** Directly managed distributed teams in: USA, Canada, UK, Germany, Poland, India, and Singapore.